

8

Pump Station Design

In this chapter we will look at some of the significant items in a pump station that pertain to pumps and pipeline hydraulics. We will analyze the various pressures on both the suction and discharge side of the pump and how pressure control is implemented using a control valve downstream of the pumps. The amount of pump throttle pressure due to mismatch between pump head and system curve head will be analyzed and the amount of horsepower wasted will be calculated. The use of variable speed drive (VSD) pumps to eliminate throttle pressure by providing just enough pressure for the specified flow rate is explained. We also perform a simple economic comparison between a VSD pump installation and a constant-speed pump with a control valve.

8.1 Suction Pressure and Discharge Pressure

A typical piping layout within a pump station is as shown in [Figure 8.1](#). The pipeline enters the station boundary at point A, where the station block valve MOV-101 is located. The pipeline leaves the station boundary on the discharge side of the pump station at point B, where the station block valve MOV-102 is located.

Station bypass valves designated as MOV-103 and MOV-104 are used for bypassing the pump station in the event of pump station maintenance

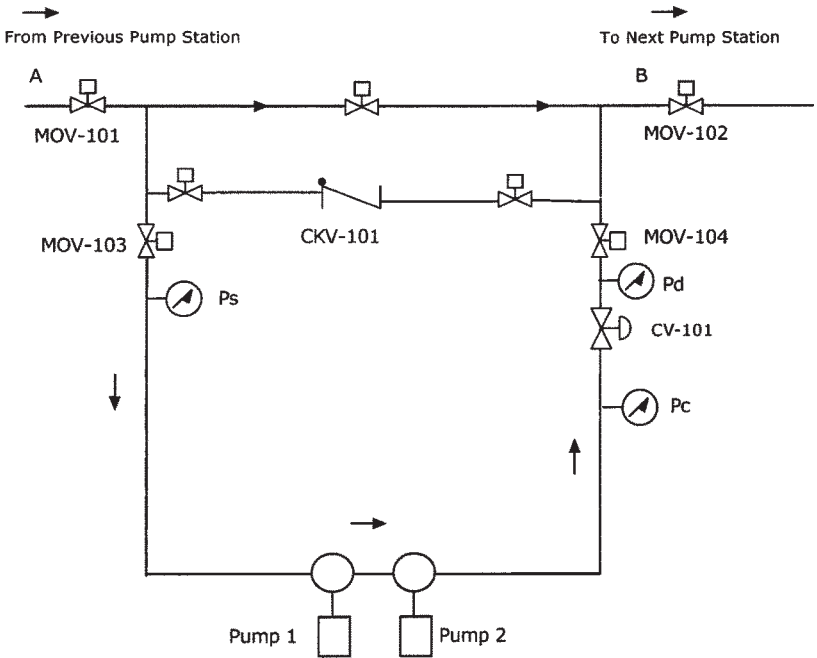


Figure 8.1 Typical pump station layout.

or other reasons when the pump station must be isolated from the pipeline. Along the main pipeline there is located a check valve, CKV-101, that prevents reverse flow through the pipeline. This typical station layout shows two pumps configured in series. Each pump pumps the same flow rate and the total pressure generated is the sum of the pressures developed by each pump. On the suction side of the pump station the pressure is designated as P_s while the discharge pressure on the pipeline side is designated as P_d . With constant-speed motor-driven pumps, there is always a control valve on the discharge side of the pump station, shown as CV-101 in Figure 8.1. This control valve controls the pressure to the required value P_d by creating a pressure drop across it between the pump discharge pressure P_c and the station discharge pressure P_d . Since the pressure within the case of the second pump represents the sum of the suction pressure and the total pressure generated by both pumps, it is referred to as the case pressure P_c .

If the pump is driven by a VSD motor or an engine, the control valve is not needed as the pump may be slowed down or speeded up as required to

generate the exact pressure P_d . In such a situation, the case pressure will equal the station discharge pressure P_d .

In addition to the valves shown, there will be additional valves on the suction and discharge of the pumps. Also, not shown in [Figure 8.1](#), is a check valve located immediately after the pump discharge that prevents reverse flow through the pumps.

8.2 Control Pressure and Throttle Pressure

Mathematically, if ΔP_1 and ΔP_2 represent the differential head produced by pump 1 and pump 2 in series, we can write

$$P_c = P_s + \Delta P_1 + \Delta P_2 \quad (8.1)$$

where

P_c = Case pressure in pump 2 or upstream pressure at control valve The pressure throttled across the control valve is defined as

$$P_{thr} = P_c - P_d \quad (8.2)$$

where

P_{thr} = Control valve throttle pressure

P_d = Pump station discharge pressure

The throttle pressure represents the mismatch that exists between the pump and the system pressure requirements at a particular flow rate. P_d is the pressure at the pump station discharge needed to transport liquid to the next pump station or delivery terminus based on pipe length, diameter, elevation profile, and liquid properties. The case pressure P_c , on the other hand, is the available pressure due to the pumps. If the pumps were driven by variable-speed motors, P_c would exactly match P_d and there would be no throttle pressure. In other words, the control valve would be wide open and obviously would not be needed in operation. The case pressure is also referred to as control pressure since it is the pressure upstream of the control valve. The control valve also functions as a means for protecting the discharge piping. The throttle pressure represents unused pressure developed by the pump and hence results in wasted horsepower (HP) and money. The objective should be to reduce the amount of throttle pressure in any pumping situation.

As mentioned earlier, VSD pumps can speed up or slow down to match pipe pressure requirements. In such situations there is no control valve and therefore the throttle pressure is zero. With VSD pumps there is no HP wasted since the pump case pressure exactly matches the station discharge pressure.

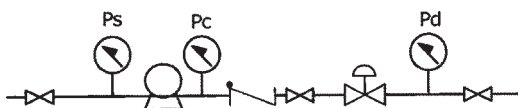


Figure 8.2 Single pump schematic.

A simplified pump station schematic with one pump is shown in Figure 8.2.

Example Problem 8.1

A pump station installation at Corona consists of two pumps configured in series each developing 1500 ft of head at 4000 gal/min. Diesel fuel ($\text{Spgr}=0.85$, and $\text{viscosity}=5.9 \text{ cSt}$) is pumped from Corona to a delivery terminal at Sunnymead, 75 miles away. The required discharge pressure at Corona based on pipe size, pipe length, liquid properties, elevation profile between Corona and Sunnymead, and the required delivery pressure at Sunnymead has been calculated to be 1050 psi. The pump station suction pressure is maintained at 50 psi to prevent pump cavitation. Assume the combined efficiency of both pumps at Corona at the given flow rate to be 82%.

- Analyze the pump station pressures and determine the amount of throttle pressure and HP wasted.
- If electrical energy costs 8 cents per kWh, estimate the dollars lost in control valve throttling.
- What recommendation can you make to improve pipeline operation?

Solution

- The total pump pressure developed by two pumps in series is

$$\frac{(1500 + 1500) \times 0.85}{2.31} = 1104 \text{ psi, rounded off}$$

Adding the 50 psi suction pressure gives a control pressure upstream of the control valve of

$$1104 + 50 = 1154 \text{ psi}$$

This is also the case pressure in the second pump. Since the station discharge pressure is 1050 psi, the control valve throttle pressure is

$$\text{Throttle pressure} = 1154 - 1050 = 104 \text{ psi}$$

The HP wasted due to pump throttling can be calculated as

$$\text{HP wasted} = \frac{\text{gal/min} \times \text{psi}}{1714 \times \text{efficiency}} \quad \text{using Equation}$$

or

$$\text{HP wasted} = \frac{4000 \times 104}{1714 \times 0.82} = 296$$

(b) At 8 cents per kWh electric cost, above wasted HP is equivalent to

$$296 \times 0.746 \times 24 \times 350 \times 0.08 = \$148,388 \text{ per year}$$

assuming 350 days of continuous operation per year. This is a substantial loss.

(c) If plans are to continue operating this pump station at the current flow rate on a continuous basis, it would be preferable to trim the pump impellers to eliminate the throttle pressure and hence wasted HP. A spare rotating element for the pump can easily be purchased and installed for about \$50,000—only about a third of the money wasted annually in throttling. Therefore, the recommendation is to purchase and install a new trimmed rotating element for one of the pumps that will result in no throttling at 4000 gal/min. The existing larger impeller may be stored as a spare for future use when increased flow rates are anticipated.

8.3 Variable-Speed Pumps

It was mentioned in the preceding section that when pumps are driven by variable-speed motors or gas turbine drives, the control valve is not needed and the pump throttle pressures will be zero. The variable-speed pump will be able to speed up or slow down to match the pipeline pressure requirements at any flow rate, thereby removing the need for a control valve. Of course, depending upon the pump, there will be a minimum and maximum permissible pump speeds.

If there are two or more pumps in series configuration, one of the pumps may be driven by a VSD motor or an engine. With parallel pumps, all pumps will have to be VSD pumps, since parallel configuration requires matching heads at the same flow rate.

In the case of two pumps in series, similar to Example Problem 8.1, we could convert one of the two pumps to be driven by a variable-speed motor. This pump can then slow down to the required speed that would develop just the right amount of head (at 4000 gal/min) which, when added to the

1500 ft developed by the constant-speed pump, would provide exactly the total head needed to match the pipeline system requirement of 1050 psi.

Example Problem 8.2

Use the data in Example Problem 8.1 and consider one of the two pumps driven by a variable-speed electric motor. Calculate the speed at which the VSD pump should be operated to match the pipeline requirements at 4000 gal/min. Assume that the rated speed for the pump at 3560 RPM produces a head of 1500 ft at 4000 gal/min.

Solution

The pipeline pressure requirement is given as 1050 psi. Assuming the VSD pump develops a head of H ft, the total head produced by the two pumps in series is $(H+1500)$ ft. Since the pump suction pressure is 50 psi, the total pressure on the discharge side of the pumps is

$$(H+1500) \times 0.85/2.31 + 50 \text{ psi}$$

This must exactly equal the discharge pressure requirement of 1050 psi at the 4000 gal/min flow rate. Therefore, we can state that

$$1050 = 50 + (H+1500) \times 0.85/2.31$$

From which, we get

$$H = 1218 \text{ ft}$$

This is the head that the VSD pump must develop, compared with 1500 ft for the constant-speed pump.

At the rated speed of 3560 RPM, this pump develops 1500 ft at 4000 gal/min. In order to reduce the head to 1218 ft, the VSD pump has to be slowed down to some speed N RPM. We will calculate this speed using Affinity Laws.

Since head is proportional to the square of the speed,

$$\text{Speed ratio} = N/3560 = (1218/1500)^{1/2} \text{ using Equation (7.8)}$$

Solving for N we get

$$N = 3208 \text{ RPM approximately}$$

By operating the second pump (as a VSD pump) at 3208 RPM in series with the first pump (constant speed) the pump station will produce exactly 1050 psi required to pump the diesel fuel at 4000 gal/min. Therefore, no pump throttling will be required and no pump BHP will be wasted.

8.3.1 VSD Pump Versus Control Valve

In a single pump station pipeline with one pump used to provide the pressure required to pump the liquid, a control valve is used to regulate the pressure for a given flow rate.

Suppose the pipeline from the Essex pump station to the Kent delivery terminal is 120 miles long and is constructed of a 16 in. diameter and 0.250 in. wall thickness pipe with a maximum allowable operating pressure (MAOP) of 1440 psi (Figure 8.3). The pipeline is designed to operate at 1400 psi, pumping 4000 bbl/hr of liquid (specific gravity 0.89 and viscosity 30 cSt at 60°F) on a continuous basis. The delivery pressure required at Kent is 50 psi. If the pump suction pressure at Essex is 50 psi, the required pump differential pressure is 1350 psi. Unless the single pump unit at Essex was specifically selected for this application, the chances are that the Essex pump H-Q curve may indicate the following:

$$Q=2800 \text{ gal/min } H=3800 \text{ ft}$$

The flow rate of 2800 gal/min is the equivalent of 4000 bbl/hr. Converting the head available (3800 ft) into psi, the pump pressure developed is

$$3800 \times 0.89 / 2.31 = 1464 \text{ psi}$$

On the other hand, if the pump head at $Q=2800 \text{ gal/min}$ were only 3200 ft, the pressure developed would only be

$$3200 \times 0.89 / 2.31 = 1233 \text{ psi}$$

which is inadequate for our application that requires a pressure of 1350 psi to pump the liquid at 4000 bbl/hr.

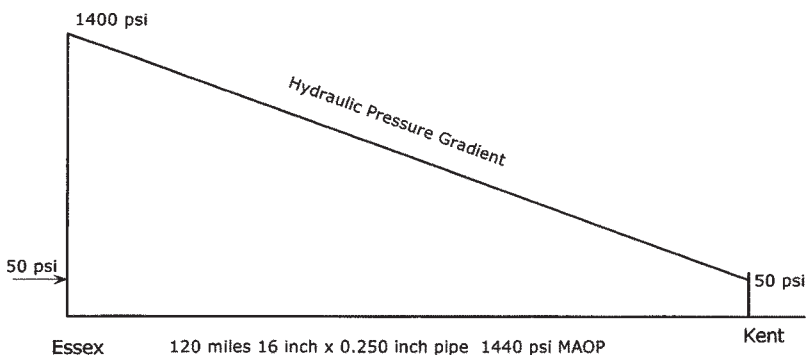


Figure 8.3 Essex to Kent pipeline.

Assuming the first case where the pump is able to develop 3800 ft that corresponds to 1464 psi, we can see that this pressure combined with the 50 psi pump suction pressure would produce a pump discharge pressure (actually the pump case pressure) of

$$1464 + 50 = 1514 \text{ psi}$$

As mentioned earlier, the pipeline pressure limit (MAOP) is 1440 psi; as a result we will over-pressure the pipeline by 74 psi. Obviously, this cannot be tolerated and we must have some means to control the pump pressure to the required 1400 psi pump station discharge. A control valve located downstream of the pump discharge will be used to reduce the discharge pressure to the required pressure of 1400 psi. This is depicted by the modified system curve (2) in Figure 8.4.

The system head curve (1) in Figure 8.4 represents the pressure versus flow rate variation for our pipeline from Essex to Kent. At $Q = 2800$ gal/min (4000 bbl/hr) flow rate, point C on the pipeline system head curve shows the desired operating point that requires a pipeline discharge pressure of 1400 psi at Essex. Since we have superimposed the pump H-Q curve and pressures are in ft of liquid head, the pressure at C is

$$1400 \times 2.31 / 0.89 = 3634 \text{ ft}$$

Since the pump suction pressure is 50 psi, we have plotted the pump curve in Figure 8.4 to include this suction head as well. The pump H-Q curve

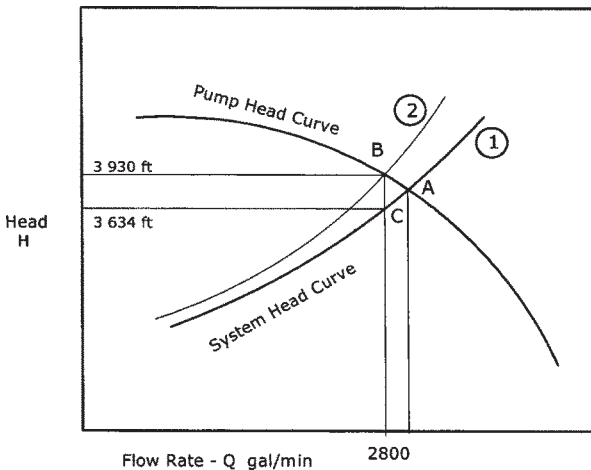


Figure 8.4 System curve and control valve.

shows that at a flow rate of 2800 gal/min, the differential head generated is 3800 ft. Adding the 50 psi suction pressure, point B on the H-Q curve corresponds to

$$3800 + 50 \times 2.31 / 0.89 = 3930 \text{ ft}$$

We can see from the point of intersection of the pump H-Q curve and the pipeline system curve that the operating point will be at point A, which is a higher flow rate than 2800 gal/min. Also, the pump discharge pressure at A will be at a higher flow rate than that at C. Unless the pipe MAOP is greater than the pressure at A, we cannot allow this pump to operate without pressure control. Thus utilizing the control valve on the pump discharge we will move the operating point from A to B on the pump curve corresponding to 2800 gal/min and a higher head of 3930 ft as calculated above. The control valve will then cause the pressure drop equivalent to the head difference BC calculated as follows:

$$\text{Control valve pressure drop, } BC = 3930 - 3634 = 296 \text{ ft}$$

This pressure drop across the control valve is also called the pump throttle pressure:

$$\text{Throttle pressure} = 296 \times 0.89 / 2.31 = 114 \text{ psi}$$

We thus conclude that, due to the slightly oversized pump and the pipe MAOP limit, we must utilize a control valve on the pump discharge to limit the pipeline discharge pressure to 1400 psi at the required flow rate of 4000 bbl/hr (2800 gal/min). A hypothetical system head curve, designated as (2), passing through point B on the pump curve, is the artificial system head curve due to the restriction imposed by the control valve.

It must be pointed out that as far as the pump is concerned, the suction pressure, case pressure, discharge pressure, and throttle pressure are as follows

$$\text{Pump suction pressure} = 50 \text{ psi}$$

$$\text{Pump case pressure} = 1514 \text{ psi}$$

$$\text{Pump station discharge pressure} = 1400 \text{ psi}$$

$$\text{Pump throttle pressure} = 114 \text{ psi}$$

The above analysis applies to a pump driven by a constant-speed electric motor. If we had a VSD pump that can vary the pump speed from 60% to 100% rated speed and if the rated speed were 3560 RPM, the pump speed range would be

$$3560 \times 0.60 = 2136 \text{ RPM to } 3560 \text{ RPM}$$

Since the pump speed can be varied from 2136 RPM to 3560 RPM, the pump head curve will correspondingly vary according to the centrifugal pump Affinity Laws. Therefore we can find some speed (less than 3560 RPM) at which the pump will generate the required head corresponding to point C in [Figure 8.4](#).

Point C represents a pump differential head of

$$(1400-50) \times 2.31 / 0.89 = 3504 \text{ ft}$$

where 50 psi represents the pump suction pressure. If the given H-Q curve is based on the rated speed of 3560 RPM, using Affinity Laws we calculate the approximate speed required for point C as

$$3560(3504/3800)^{1/2} = 3419 \text{ RPM}$$

If we had the full H-Q curve data at 3560 RPM we could generate the revised H-Q curve at the desired speed of 3419 RPM. Actually, we may have to adjust the speed a little to get the required head corresponding to point C.

From the foregoing analysis we can conclude that use of VSD pumps can provide the right amount of pressure required for a given pump flow rate, thus avoiding the pump throttle pressures (and hence wasted HP) that are common with constant-speed motor-driven pumps with control valves. However, VSD pumps are expensive to install and operate compared with the use of a control valve. A typical control valve installation may cost \$100,000, whereas a VSD may require \$300,000 to \$500,000 incremental cost compared with a constant-speed motor-driven pump. We will have to factor in the increased operating cost of the VSD pump compared to the money lost in wasted HP from control valve throttling.

Let us illustrate this function by considering the Essex to Kent pipeline again. In the first case, we will assume a constant-speed pump at Essex with a control valve. The pump and control valve installation will cost approximately

$$$(X+100,000)$$

where X represents the cost of the constant-speed motor-driven pump. The annual maintenance cost for the control valve is estimated at \$10,000. Due to pump throttling, let us assume an average pressure drop in the control valve to be 100 psi over a 12 month period at an average pipeline flow rate of 4000 bbl/hr. This is equivalent to a wasted HP of

$$\frac{2800 \times 100}{1714 \times 0.8} = 204.2 \text{ HP}$$

Assuming 80% pump efficiency, at an average electrical cost of 10 cents/kWh and 350 days operation per year, the throttled pressure represents a loss of

$$\$ (204.2 \times 0.746 \times 24 \times 350 \times 0.10) = \$127,960 \text{ per year}$$

If we now replaced the pump at Essex with a VSD pump consisting of a variable frequency drive (VFD) electric motor with the same pump without the control valve, the capital cost will be approximately

$$\$ (X + 500,000)$$

where \$500,000 has been added to account for the additional switchgear and the VFD motor. The annual maintenance cost for this installation may be as high as \$50,000.

Comparing the two alternatives, we can summarize as follows, assuming $X = 250,000$ for the pump motor.

Case A: Constant-speed motor-driven pump with control valve

Capital cost \$350,000

Annual cost \$137,960

(includes HP lost due to control valve)

Case B: VFD motor-driven pump, no control valve

Capital cost \$750,000

Annual cost \$50,000

It is clear that the VSD pump has higher initial cost but lower annual cost compared with the constant-speed motor/control valve combination.

In this example, we can perform an economic analysis and compare case A with case B considering discounted cash flow. If the interest rate is 8% per year the present value of case A for a 15 year project life is

$$PV_A = 350,000 + PV(137,960, 15, 8)$$

where $PV(A, N, i)$ represents the present value of a series of cash flows of \$A, each for N years at an annual interest rate of i percent. Therefore

$$\begin{aligned} PV_A &= 350,000 + 1,180,866 \\ &= \$1,530,866 \end{aligned}$$

Similarly,

$$\begin{aligned} PV_B &= 750,000 + PV(50,000, 15, 8) \\ &= 750,000 + 427,974 \\ &= \$1,177,974 \end{aligned}$$

It can be seen from the above that the VSD case will be the more economical alternative, since it has a lower present value of investment.

The advantages of VSD pumps include operational flexibility and lower power requirements since no HP is wasted as with a control valve. The disadvantage of VSD pumps include higher capital and operating cost compared with a constant-speed pump with a control valve.

8.4 Summary

This chapter covered the main components found in a typical pump station as they relate to pumps and pipeline hydraulics. The concepts of pump suction pressure, case pressure, pump station discharge pressure, and throttle pressure were introduced and calculations illustrated using examples. The impact of pump throttle pressure and how the control valve is used to protect the discharge piping as well as to provide the necessary pressure for a particular flow rate were explained. We found that the throttle pressure contributes to energy loss and therefore wasted money. The advantages of using VSD pumps to provide just enough pressure for the specified flow, thereby eliminating throttle pressures, were analyzed. Also illustrated was a comparison of using a VSD pump versus a control valve.

8.5 Problems

- 8.5.1 The origin pump station at Harvard on a 50 mile pipeline consists of two pumps in series, each developing 1450 ft of head at 3000 bbl/hr. Diesel fuel (specific gravity 0.85 and viscosity 5.9 cSt) is pumped from Harvard to a delivery terminal at Banning. The required discharge pressure at Harvard has been calculated to be 995 psi. The pump station suction pressure is maintained at 50 psi. Use a combined pump efficiency of 84% at the given flow rate.
- (a) Analyze the pump station pressures and determine the amount of throttle pressure and HP wasted.
 - (b) If electrical energy costs 10 cents/kWh, estimate the dollars lost in control valve throttling.
- 8.5.2 If a VSD pump is used in Problem 8.5.1 above, determine the speed at which the pump should be run to maintain the flow rate of 3000 bbl/hr. The rated pump speed is 3560 RPM.
- 8.5.3 A water pipeline consists of three pump stations, each having one constant-speed motor-driven pump of 2000 HP. The analysis of current operations shows that the pipeline has been operating at a fairly constant flow rate of 3000 gal/min, and the pump throttle pressures are as follows:

Pump station 2:75 psi

Pump station 3:85 psi

Pump station 1 has zero throttle pressure. Trimming the pump impellers to the correct size so as to eliminate wasted HP from pump throttle is estimated to cost \$50,000 per pump station. Retrofitting each pump station with VFD motor-driven pumps is estimated to cost \$500,000 per pump station. The operating cost for the VFD units is \$50,000 per site.

- (a) Determine the economics of installing trimmed impellers and the payment period based on 8% interest rate. Assume 80% pump efficiency for HP calculations and an electricity cost of 10 cents/kWh.
- (b) If it was decided not to trim the pump impellers, compare the VFD motor option with the constant-speed pump option. Consider the control valve annual operating cost to be \$5000 per site.

8.5.4 A refined products pipeline is used for batched operations with gasoline, kerosene and diesel fuel. The pipeline is 80 miles long and is 14 in. in diameter, 0.250 in. wall thickness. The MAOP of the pipeline is 1200 psi.

- (a) Determine the maximum throughput possible with one pump station for each product considering each product pumped alone with no batching. Assume a flat elevation profile with 50 psi pipe delivery pressure. For liquid properties use the following:

Product	Specific gravity	Viscosity (cSt)
Gasoline	0.74	0.6
Kerosene	0.82	2.0
Diesel	0.85	5.5

- (b) Investigate the use of VSD pumps to provide the required operational control and flexibility when this pipeline is operated in a batched mode. Pick a typical pump curve for the origin pump station to first satisfy the need to maximize gasoline throughput. Having based pump requirements on

gasoline, determine the speed variants required for the other two products without exceeding the MAOP.

- (c) Estimate the annual power cost for each product based on an electricity cost of 10 cents/kWh.
- (d) How would the operating scenario change if VSD pumps were not used? Estimate the cost of HP wasted in pump throttling with the control valve.